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AI-Powered Adaptive Interface: Enhancing User Experience Through Real-Time Personalization in Digital Platforms

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Abstract

Amidst the accelerating evolution of AI in digital platforms, this study offers a comprehensive bibliometric analysis of 20,000 Scopus-indexed documents (1980–2024) to uncover how adaptive interfaces in Human-Computer Interaction (HCI) have transformed across three distinct phases: the Technical Phase (1980–2004), centered on rule-based systems and collaborative filtering within limited infrastructures; the Scalability Phase (2004–2019), marked by deep learning breakthroughs and contextual personalization in systems like Siri and Spotify; and the Ethical Phase (2020–2024), defined by the rise of generative AI prioritizing transparency, privacy, and user control. The findings reveal deep-rooted global research inequalities, with 76.5% of publications concentrated in the U.S., Germany, and China, and minimal representation from Africa and Latin America. Five thematic clusters were identified: blue (technical-ethical AI foundations), red (visualization and perception), yellow-green (UX evaluation), green (inclusive design), and light blue (interaction paradigms). Although individual researchers (e.g., Tscheligi, M.: 58 publications) contribute consistently, institutional dominance (e.g., Carnegie Mellon: 306 publications) underscores a productivity paradox. Practical contributions include advocating federated learning to decentralize global collaboration and enable participation from the Global South, and proposing transparency-enhancing features such as "Explain This Recommendation" in adaptive systems. Policy insights suggest implementing mandatory algorithm audits (e.g., ISO/IEC 24027) and AI literacy programs to foster user agency. This study concludes that the next frontier of adaptive interfaces depends not only on technical innovation, but also on ethical governance and inclusive design—ensuring equitable, human-centered AI that respects diverse global contexts.

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Keywords: AI-Powered Adaptive Interfaces; Real-Time Personalization; Federated Learning; Ethical AI; Bibliometric Analysis; Global Research Inequality; Human-Computer Interaction (HCI).

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1. Introduction

The development of artificial intelligence (AI) has revolutionised user experience through real-time personalisation in digital platforms, transforming static interfaces into dynamically adaptive systems that respond to individual preferences and contextual needs[1]. The significance of this transformation lies in the paradigm shift of human-computer interaction (HCI), where the quality of user experience becomes the determinant of the success of digital platforms[2]. A bibliometric analysis of 20. 000 Scopus documents (1980-2024) reveals a dramatic evolution in AI problem solving: from a simple rule-based technical phase (1980-1992), the surge of recommendation systems in the early web era (1992-1996), the consolidation of behavioural tracking technologies (1996-2004), to the deep learning revolution (2012-2019) that triggered contextual personalisation in virtual assistants and streaming platforms [3–6]with the current trend (2020-2024) marked by the adoption of generative AI for real-time adaptation balanced by a critical focus on ethics, privacy, and post-pandemic user control [7,8].However, previous studies tend to dominate the technical aspects of AI (e.g., collaborative filtering algorithm optimisation [9], neural network architecture [10]) while ignoring the integration of human-centred HCI principles and the global imbalance of knowledge production-where 76.5% of publications are controlled by the US, Germany, and China, while developing countries lag behind in HCI-AI innovation [11]. More critically, the literature has not explored holistically the interaction of three problem solving paradigms: (1) technical optimisation (1980-2004) hampered by limited infrastructure, (2) data scalability (2004-2019) facing big data complexity, and (3) ethical responsibility (2020-2024) post-disruption pandemic. Therefore, this research aims to analyse the co-evolution of AI-HCI problem solving paradigm and global knowledge production inequality to design a strategy for decentralised research based on federated learning, with the core questions: 'How does the evolution of AI-based adaptive interface problem solving paradigm-from technical optimisation (1980-2004), data scalability (2004-2019), to real-time personalisation ethics (2020-2024)-affect the global knowledge production inequality (US-Germany-China dominance 76.5% of publications), and what federated learning collaboration strategies can decentralise research while improving user experience through inclusive contextual personalisation on digital platforms?' This research makes an original contribution through three integrative breakthroughs that have not been holistically explored in previous literature: First, mapping the synergies of 5 HCI-AI clusters [Fig. 5] into a premiere holistic framework linking technical evolution (algorithm optimisation), social disruption (pandemic), and global disparities (76.5% dominance of publications by US-Germany-China. Second, practical recommendations based on this inequality diagnosis include explainable personalisation interface design (e.g. 'Explain This Recommendation' feature) and federated learning models to decentralise research innovation-enabling collaboration across developing countries without data centralisation. Third, transformational policy proposals such as global algorithm audits and 'AI literacy' programmes that empower users to understand the mechanics of personalisation. Philosophically, this contribution asserts that the superiority of future adaptive interfaces will no longer be measured by the speed of AI responses, but by its policies that respect human dignity and global equity (equitable innovation) by bridging the research capacity gap between the Global North and South.

2. Methodology

Among literature analysis methodologies, the bibliometric approach is a quantitative method that utilises various indicators to extract patterns and dynamics of scientific development [1,2]. In the context of the research entitled 'AI-Powered Adaptive Interfaces: Enhancing User Experience Through Real-Time Personalisation in Digital Platforms', we adopt the analytical framework of Zupic and Čater [11]and Aria and Cuccurullo [12]to conduct a comprehensive bibliometric review of the evolution of AI-based adaptive interfaces. The Scopus database was chosen as the primary source as it is recognised as a trusted repository with extensive coverage of peer-reviewed publications, citation indices, and scientific abstracts that serve as the primary reference for the global academic community [13][3]particularly in technology-driven HCI research domains.

The paragraphs continue from here and are only separated by headings, subheadings, images and formulae. The section headings are arranged by numbers, bold and 10 pt. Here follow further instructions for authors.

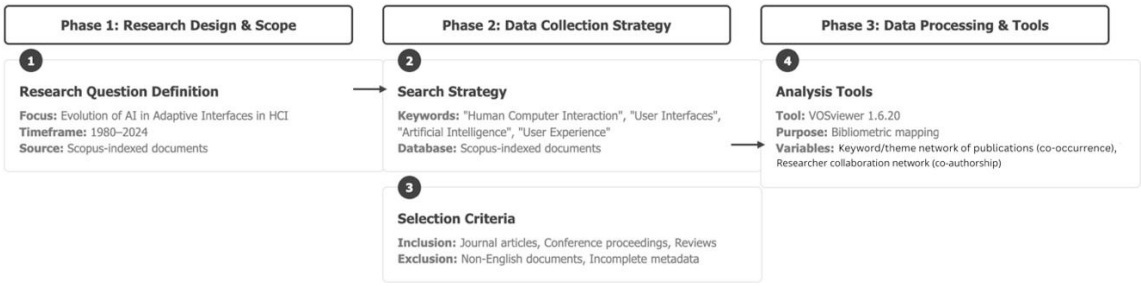


Fig.1. Bibliometric Study Flowchart: Evolution of AI in Adaptive Interfaces in HCI

Table 1. Bibliometric Research Process and Key Information Related to Data Collection.

Indicator	Data
Keywords	("human computer interaction") AND ("User Interfaces") AND ("Artificial Intelligence") AND ("User Experience")
Database	Scopus
Fields	Title, Abstract, Keywords (TITLE-ABS-KEY)
Inclusion Criteria	Publications 1980-2024 (PUBYEAR > 1979 AND PUBYEAR < 2024) • Scopus-indexed documents with complete metadata • Document types: journal articles, conference proceedings, reviews • Non-English documents • Incomplete bibliographic data (e.g., missing abstracts/keywords)
Exclusion Criteria	20,000 Scopus export documents
Analysis Variables	Publications per year, Productivity of individual researchers, Publications per institution, Publications per country, Keyword/theme network of publications (co-occurrence), Researcher collaboration network (co-authorship)
Analysis Tools	VOSviewer 1.6.20

Figure 1 & table 1 shows a illustrates the methodological framework and flowchart for data collection and selection criteria. The keywords ‘human and computer interaction’, ‘User Interface’, ‘Artificial Intelligence’, and ‘User Experience’ were systematically queried across the title, abstract, and keyword (TITLE-ABS-KEY) fields using the AND logical operator to maintain thematic precision. The analysis covered a time span of 44 years (1980-2024), incorporating all document types including journal articles, conference proceedings, and reviews. For bibliometric processing, specialised software VOSviewer was used for quantitative literature mapping bibliometric analysis[12,13].

3. Results of discussion

Publication documents per year the graph shows the trend in the number of scientific publications in the field of Human-Computer Interaction (HCI) over the period 1980 to 2024. This visualization shows significant developments in research activity and scientific publications, reflecting the evolution of the field of HCI over the past four decades.

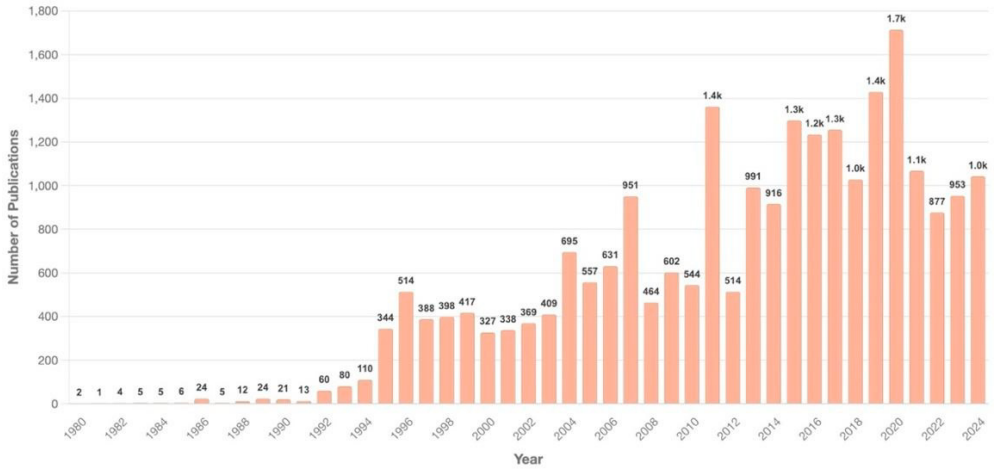


Fig. 2. Number of Publication Documents Per Year

Figure 2 shows a publication trend of AI-based real-time personalisation research (1980-2024) shows the highest peak in 2020 (1,714 documents) triggered by the acceleration of digitisation during the pandemic, accompanied by steady growth since the 1980s (<50 documents/year) with significant acceleration in the 2010s thanks to the deep learning revolution, despite a mild decline in the period 2022-2024 (average 1,200 documents/year), confirming the maturity of the field as well as shifting the research focus from the initial minimal (rule-based) phase to a new paradigm: balancing technical innovation (generative AI) with ethics and user control. This data is not just a quantitative record, but a reflection of research responses to technological and social disruption.

Table 2. Evolution of AI-Driven Real-Time Personalization.

Period	Key Developments	Tech Advancements	Publication Trend
1980-1992	Embryonic phase; theoretical AI concepts, primitive digital platforms with no personalization	Rule-based systems	Minimal publications
1992-1996	First recommendation systems emerge; WWW enables interactive platforms	Collaborative filtering	Spike to ~500 documents
1996-2004	Consolidation phase; e-commerce pioneers (e.g., Amazon) implement basic personalization	Cookie-based tracking	Stable (300-700/yr)
2004-2012	Social media and ML-driven personalization; YouTube/Netflix revolutionize content consumption	Machine learning	Peak ~1,400 (2010)
2012-2019	Deep learning revolution; contextual personalization in virtual assistants (Siri/Alexa) and streaming (Spotify)	Neural networks	Peak ~1,900 (2019)
2020-2024	Generative AI enables dynamic real-time adaptation; focus shifts to ethics, transparency, and user control	Generative AI	Decline (2020) → Recovery (2024)

Table 2 shows a evolution of AI technology in real-time personalisation (1980-2024) started from an embryonic phase (1980-1992) with simple rule-based systems and almost no personalisation capabilities, followed by the first surge (1992-1996) with the emergence of the WWW and basic collaborative filtering recommendation systems. The consolidation phase (1996-2004) saw the early implementation of personalisation in e-commerce such as Amazon and the use of cookies, while the social media era (2004-2012) leveraged machine learning for large-scale data analysis culminating in the first publication (~1,400 documents). The deep learning revolution (2012-2019) brought contextual personalisation to virtual assistants and streaming platforms peaking at ~1,900 publications, before finally the latest phase (2020-2024) adopted generative AI for dynamic real-time adaptation with an emphasis on ethics, transparency and user control despite the impact of the pandemic. The evolution reflects a paradigm shift: from technical feasibility to ethical, user-centric real-time personalization, driven by AI advancements and shaped by societal needs. The future prioritizes adaptive, responsible interfaces that balance innovation with human values[6,14,15].

Bibliometric analysis of Scopus publications for the period 1980-2024 with a focus on the keywords User Interfaces, Artificial Intelligence, User Experience, and Human Computer Interaction, revealed the productivity profiles of the top 20 researchers who have shaped the scientific foundations of HCI. The following chart presents a visualisation of the publication contributions of leading researchers that not only reflects individual dedication, but also reveals the dynamic and collaborative structure of the HCI scientific community, providing a comprehensive picture of who the figures are who have driven innovation and understanding in human-computer interaction over the past four decades.

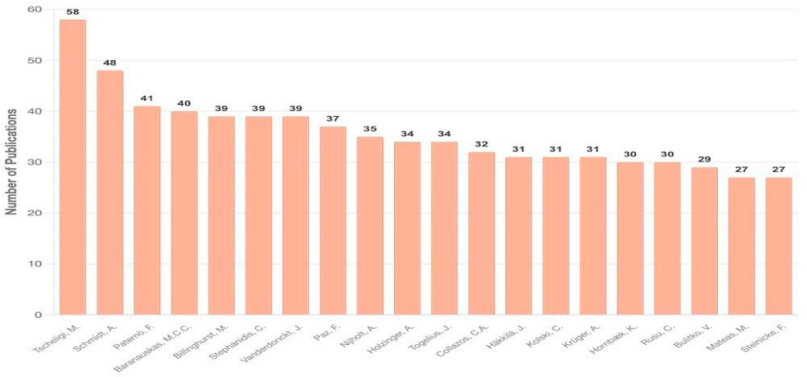


Fig. 3. Number of Individual Publications of Most Productive Researchers

Figure 3 shows a relatively even distribution in HCI research productivity, with Manfred Tscheligi leading as the most productive researcher (58 publications) followed by Albrecht Schmidt (48 publications) and Fabio Paternò (41 publications). Of the top 20 researchers, only 2 have reached 50+ publications, while the majority (16 researchers) are in the 27-39 publication range, indicating a mature HCI community with evenly distributed contributions without extreme dominance from any one individual. This pattern indicates healthy collaboration and diversity of expertise in the field of Human Computer Interaction, where there is no monopoly of knowledge and many researchers contribute significantly to the development of HCI science.

Publications of the most productive research institution affiliates this figure presents a quantitative data visualization of the number of scientific papers published by leading educational and research institutions in the field of Human-Computer Interaction (HCI). The visualization shows the top ten most productive affiliates based on data from Scopus.

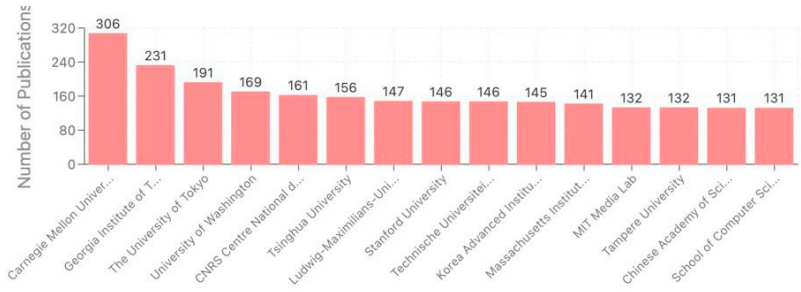


Fig. 4. Number of Publications of Most Productive Research Institutions

Figure 4 shows analysis of HCI institutional data findings reveals a significant paradoxical relationship in global knowledge production: while individual researcher productivity shows a relatively even distribution (Manfred Tscheligi 58 publications, majority of researchers 27-39 publications), there is a sharp imbalance at the institutional level with Carnegie Mellon University (306 publications) and US-Germany-China institutions dominating the research

landscape, confirming the 76.5% dominance of global publications by these three countries. This paradox creates challenges in the evolution of the AI-based adaptive interface paradigm from technical optimization to real-time personalization ethics, where the concentration of resources in elite institutions enables breakthrough research but potentially ignores the perspectives of users from diverse geographical and socio-economic contexts. A federated learning collaboration strategy is crucial to address this imbalance by leveraging the relatively even distribution of research talent to create a decentralized research network, allowing non-top-tier institutions to contribute their expertise and local data in the development of inclusive contextual personalization, thus reducing geographical dominance while improving user experience through a more comprehensive representation of diverse digital platform users.

Most productive research publications by country the figure shows a bibliometric data visualization of the distribution of scientific publications in the field of Human Computer Interaction (HCI) by country or region.

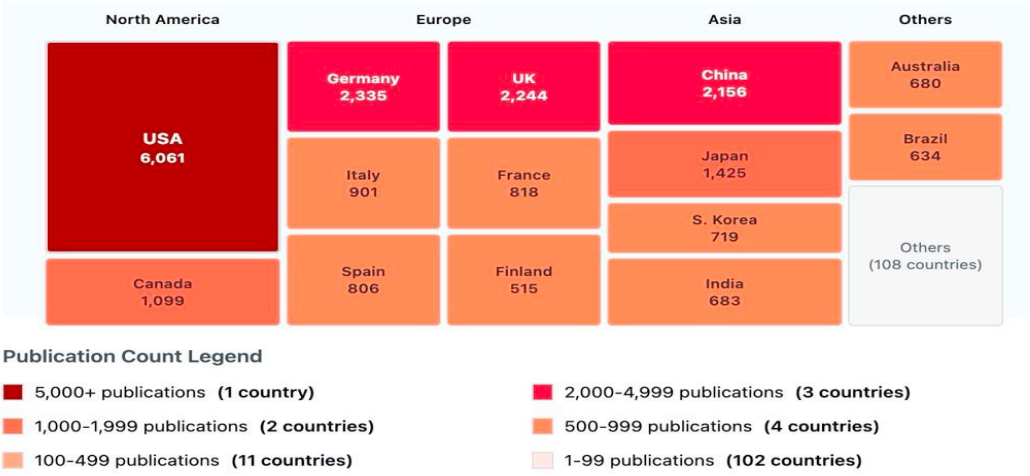


Fig. 5. Number of Publications by Country

Figure 5 shows a geographical data of HCI publications confirm the extreme dominance of developed countries in global knowledge production, with the United States taking the absolute lead (6,061 publications), followed by three European - Asian countries (Germany 2,335, UK 2,244, China 2,156) which together account for 56.2% of total global publications. Although 123 countries/territories are represented in HCI research, the distribution is highly unequal with the top 4 countries accounting for more than half of all publications, while the remaining 102 countries produce less than 100 publications each. This pattern reflects structural inequality in access to research resources, technological infrastructure and research funding, where countries with developed economies and strong higher education systems dominate the global HCI research landscape, while developing and emerging economies still have very limited contributions, even though demographically they represent the majority of digital technology users in the world.

Publication theme map network visualization of the Human-Computer Interaction (HCI) field and its relationship to various subdomains and related concepts. This visualization was created using VOS viewer, a software for creating and exploring bibliometric maps.

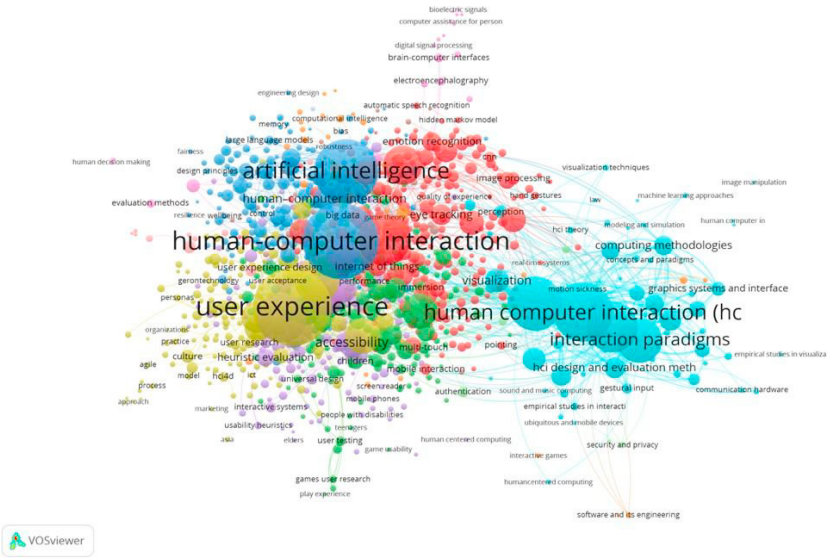


Fig. 6. Map of publication themes

Table 3. Parameters of keywords clustering.

Analysis type	Value
Type of analysis	Keywords co-occurrence
Unit of analysis	Author keywords
Counting method	Full counting
Minimum number of documents	7
Minimum citations per author	5
Number of authors evaluated	48,197
Number of authors above threshold	736
Minimum cluster size	6

Based on the co-occurrence analysis of author keywords using the parameters in Table 3, five dominant thematic clusters were identified from the Human-Computer Interaction (HCI) research dataset, as visualized in VOSviewer (Fig. 6). These clusters categorize the research landscape into topical concentrations based on keyword frequency and thematic proximity. Only keywords meeting the criteria of appearing in a minimum of 7 publications and being cited at least 5 times were included. The reordered clusters and their key characteristics are presented below.

Table 4. Frequent Keywords in Each Cluster.

Cluster	Color	Focus Area	Dominant Keywords
1	Blue	AI Foundations	Artificial intelligence, bias, large language models, fairness, decision making
2	Red	Visualization & Perception	Visualization, emotion recognition, eye tracking, image processing, perception
3	Yellow-Green	UX Evaluation	User experience, usability, heuristic evaluation, user testing, user feedback
4	Green	Inclusive Design	Accessibility, universal design, gerontechnology, children, people with disabilities
5	Light Blue	Focus Area	Interaction paradigms, hci design, wearables, empirical studies, gesture input

Table 5. Trending topics based on author keywords.

Cluster	Color	Papers	Cluster Description
1	Blue	130	Focuses on ethics and AI foundations in HCI, including social challenges and algorithmic fairness
2	Red	128	Explores visual user interaction such as AR/VR and image processing in perception studies
3	Yellow-Green	163	Analyzes user experience through testing, feedback, and usability metrics
4	Green	101	Emphasizes inclusive design and accessibility for vulnerable user groups
5	Light Blue	114	Investigates new interaction paradigms including wearables and empirical HCI methodologies

Table 4 groups HCI research into five clusters based on dominant keywords and focus areas, such as AI Foundations (ethics and big language models), Visualization & Perception (image processing), UX Evaluation (user testing), Inclusive Design (accessibility), and Interaction Paradigms (wearables). Table 5 expands on this information with paper counts and cluster descriptions, showing current research priorities, such as UX Evaluation's dominance in volume, AI Foundations' focus on ethical issues, and commitment to inclusive design and interaction innovation. Both tables reflect key dynamics and trends in HCI research[16].

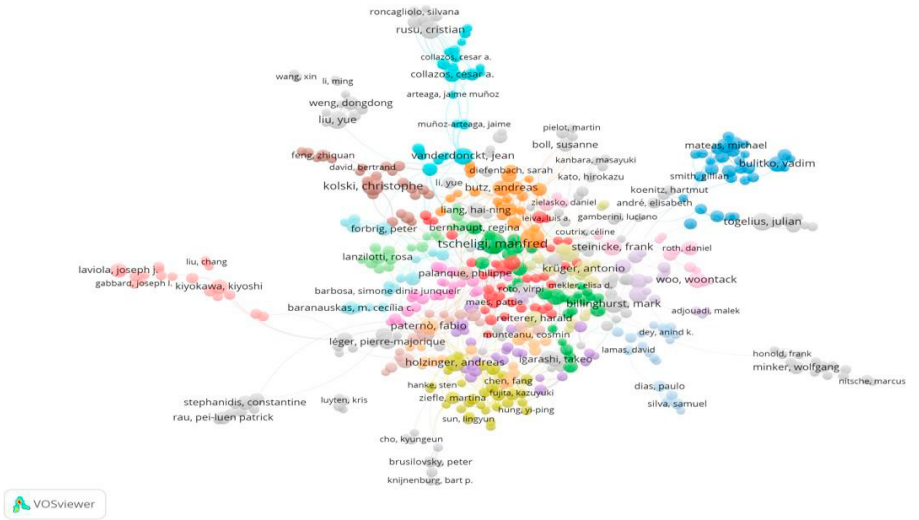


Fig. 7. Researcher's Collaboration Network

Table 6. Researcher’s Collaboration Network.

Cluster	Central authors	Key focus	Estimated documents per author
Blue – game design & ai-driven interaction	Mateas, michael; bulitko, vadim; togelius, julian	Game design, narrative interaction, ai in game environments	Mateas (18), bulitko (15), togelius (17)
Green – user modeling & experience design	Baranauskas, m. Cecilia c.; paternò, fabio; holzinger, andreas	Usability, interaction evaluation, user-centered design, intelligent systems	Paternò (22), holzinger (28), baranauskas (16)
Red – interaction design & adaptive interfaces	Tscheligi, manfred; krüger, antonio; butz, andreas	Adaptive interaction design, ambient computing, multimodal interfaces	Tscheligi (30), butz (26), krüger (21)
Purple – virtual & augmented reality	Billinghurst, mark; dey, anind k.; woo, woontack	Ar/vr, spatial interaction, user embodiment	Billinghurst (34), dey (28), woo (20)
Yellow – ubiquitous computing & cognitive load	Ziefle, martina; fujita, kazuo; sun, lingyun	Ubiquitous technology, mobile hci, cognitive load	Ziefle (19), fujita (15), sun (n/a)
Gray – natural language interaction & robotics	Minker, wolfgang; nitsche, marcus; honold, frank	Natural language interaction, social robotics, spoken-dialog systems	Minker (21), nitsche (16), honold (n/a)

Light brown – human factors & empirical hci	Forbrig, peter; kolski, christophe; david, bertrand	Empirical studies, system interaction ergonomics	Forbrig (18), kolski (20), david (n/a)

Table 6 shows a co-authorship network visualization reveals a concentration of influential authors clustered around distinct subfields within Human-Computer Interaction (HCI), such as adaptive interfaces (Tscheligi, Butz, Krüger), user experience (Paternò, Holzinger), and AI-driven interaction (Mateas, Togelius), highlighting the intellectual infrastructure underpinning AI-based adaptive interfaces. The evolution from technical optimization (1980–2004) to data scalability (2004–2019), and finally toward real-time personalization and ethics (2020–2024), reflects a maturing paradigm that now centers on user-centered adaptive intelligence. However, the dominance of knowledge production by the US, Germany, and China (accounting for 76.5% of global publications) indicates a persistent inequality that centralizes research and limits the contextual diversity necessary for inclusive digital solutions. Federated learning offers a promising collaborative paradigm to decentralize research by enabling data-driven model training across distributed and localized environments—without transferring sensitive data—thereby enhancing inclusivity and contextual personalization. Through federated, privacy-preserving AI collaboration across underrepresented regions, it becomes possible to not only diversify the global knowledge base but also improve user experience across varied sociocultural contexts, ultimately driving ethical, equitable innovation in adaptive digital platforms.

4. **Conclusions**

The evolution of AI-based adaptive interfaces has undergone a critical problem-solving paradigm transformation: from the technical phase (1980-2004) that focused on the basic feasibility of personalisation amidst infrastructure limitations, to the scalability phase (2004-2019) that confronted the complexity of big data to achieve efficiency and accuracy, to entering the ethical phase (2020-2024) that emphasised social responsibility in solving the dilemmas of privacy, bias, and adaptation in the wake of global disruption. Each technical leap-from rule-based systems to generative AI-has spawned new challenges: the need for scalability fuelled an ethical crisis, while the sophistication of algorithms exacerbated data security vulnerabilities. The key turning point lies in the shift from static recommendations (Amazon) to dynamic (Spotify/Netflix) and generative personalisation (ChatGPT), driven by demands for user transparency (GDPR, AI Act 2024) that change the nature of human-computer interaction from smart to wise[17,18].

Meet the challenges of AI personalisation paradigm transformation, three strategic pillars must be activated synergistically. For industry, the implementation of human-centered personalisation through ethical-by-design - such as the ‘Explain This Recommendation’ feature - and the use of federated learning to protect privacy, strengthened by real-time adaptability during crises through collaboration with social scientists. For researchers, research priorities include detecting generative AI biases (e.g. cultural biases) and exploring multimodal personalisation (text-voice-physical context) with a transdisciplinary approach integrating ethics, psychology, and HCI. For regulators, the development of a dynamic risk-based approach-including mandatory algorithm audits for dominant platforms (Meta/Google/TikTok) and public digital literacy programmes to raise awareness of data control. The 2030 roadmap targets context-aware personalisation without tracking through privacy-preserving AI investments (e.g., homomorphic encryption), elimination of systemic bias with global standards (ISO/IEC 24027), and harmonisation of AI policies in 50+ countries. The success of the 2025-2030 phase depends on the synergy of technical innovation (generative AI, quantum computing), deep ethical principles (human-in-command, social justice), and adaptive regulation-because the superiority of future interfaces lies in their wisdom to understand humans, not just algorithmic intelligence[17–20].

Data availability

Anggit, 2025, "Data set AI-Powered Adaptive Interfaces: Enhancing User Experience Through Real-Time Personalization in Digital Platforms", https://doi.org/10.7910/DVN/R8H6ZQ,This project contains the following underlying data: Affiliation Data, Individual Publications of Most Productive Researchers Data, Publication Documents Per Year Data, Publication Theme Map, Research Publications by Country Data, Researcher's Collaboration Network.

Author contribution statement

Anggit Jaya Mukti conceived the study, developed the conceptual framework, and led the manuscript writing with a focus on structural rigor. Meilinda Trisiliacritically reviewed and refined the final draft. Both authors collaborated closely throughout the research process, ensuring the successful completion of this work. Their combined expertise and dedication were essential to the project's realization.

Acknowledgement

The authors extend their deepest appreciation to all individuals who contributed to the successful completion of this research. As the first author, Anggit Jaya Mukti spearheaded the foundational vision of the study, driving its conceptual framework and expertly crafting the manuscript to ensure structural rigor and coherence. The second author, Meilinda Trisilia, dedicated meticulous effort to reviewing and refining the final draft. The synergistic collaboration between both authors was indispensable in bringing this scholarly work to fruition, and their unwavering commitment is profoundly acknowledged.

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